**Project Title:** Hyperfast DNS Load Balancer
**Team Name:** Ping@St.Joseph's

## 1. Problem Description

- DNS is the front door of the Internet, but traditional DNS load balancers face latency, scalability, and efficiency bottlenecks under high QPS and DDoS, impacting sub-millisecond resolution for CDNs and modern apps
- Aim: a next-gen DNS load balancer achieving sub-ms handling and resilient, intelligent steering at ≥1M QPS (simulated), integrating Anycast, real-time telemetry, and AI/ML policies
- Scope: multi-backend distribution, adaptive policies (geo, latency, health), resilience under stress, and optional encrypted DNS (DoH/DoT/DoQ) without degrading speed

## 2. Solution Proposed

- Processes ≥ 1M QPS in simulation with sub-millisecond latency.
- Distributes traffic intelligently using round-robin, weighted, latency-based, and geo-aware strategies.
- Integrates adaptive AI/ML decision-making policies for traffic steering.
- Implements real-time telemetry and health checks to detect failures and reroute quickly.
- Supports encrypted DNS protocols (DoH/DoT/DoQ) without sacrificing speed.
- Demonstrates resilience under flash crowds or simulated DDoS attacks.

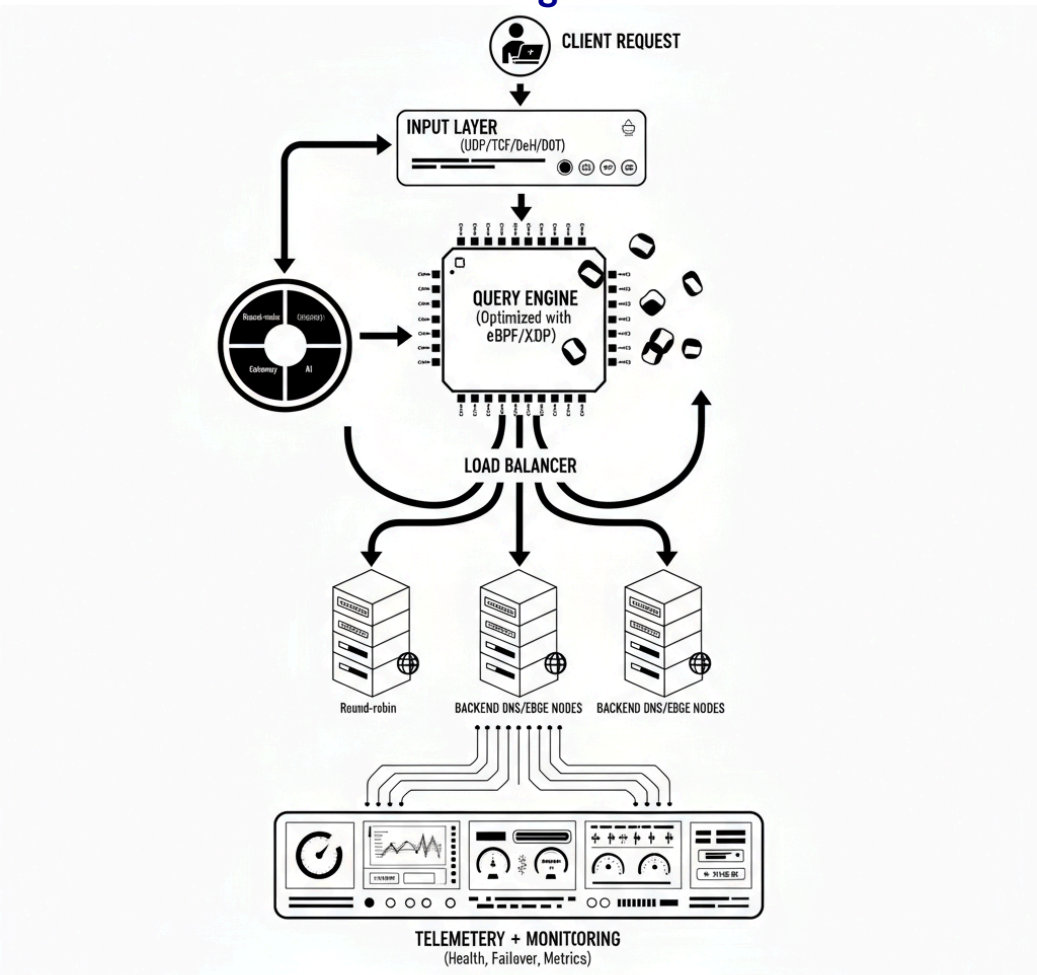
## 3. Optimization Proposed

| Area | Baseline | Optimization |
|---|---|---|
| Packet I/O | Kernel socket handling | eBPF/XDP kernel-bypass |
| Query Handling | Thread-based | Event-driven async I/O |
| Load Balancing | Round Robin | Hybrid: RR + Latency + Geo |
| Monitoring | Manual checks | AI-based anomaly detection |
| Security | Standard DNS | DoH/DoT/DoQ with TLS 1.3 |

## 4. Solution Architecture

- **Data plane**: Anycast IP → Protocol listeners (UDP/TCP/DoH/DoT/DoQ) → Fast-path parser/engine (DPDK/eBPF/XDP) → Policy engine (RR/Weighted/Latency/Geo + ML) → Backend selector → Response
- **Control plane**: telemetry collectors, health checkers, config and ML services push dynamic weights/routes with hot reload; observability via dashboards and alerts
- **Benchmarking harness**: Flamethrower load generation and latency histograms; compare with NGINX DNS/PowerDNS recursor across normal/failover/DDoS scenarios

### Design




## 5. Timeline of Delivery

| | |
|---|---|
| Sep 24–30 | Repo setup + research |
| Oct 1–7 | Build prototype DNS engine |
| Oct 8–14 | Implement LB strategies |
| Oct 15–20 | AI module + encrypted DNS |
| Oct 21–23 | Stress testing + benchmarking |
| Oct 24 | Final PDF + demo |

## 6. References


- DNS-BASED LOADBALANCING FOR WEBSERVICES
- Performance evaluation of DNS servers to build a benchmarking system of DNS64 implementations | Telecommunication Systems
- HPSRouter: A High Performance Software Router Based on DPDK
- How Fast Can a SmartNIC-Based DNS Load Balancer Run